Effat University

Lab 2: User Persona Development

FALL 2026 – CS3151

ENGINEERING COLLEGE – COMPUTER SCIENCE DEPARTMENT

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COURSE NAME: Software Engineering	COURSE NUMBER: CS3151	SECTION: 1, 2, 3

Lab Objectives:

- Identify target users for the software project
- Develop realistic user personas
- Map personas to software requirements
- Apply user-centered design thinking

Part 1: Identify User Groups

Task: List all potential user groups of your system.

Deliverable:

User Group	Role	Frequency of Use	Technical Level
School Principle	Manager	Daiy	High – Meduim
Teacher	Take attendance, enter grades, post assingnments	Daily	Meduim
Student	View timetable , grades and solve assingnmets	Daily	Meduim
Parent	Trac child`s progress,attendance and fees	Daily	Low
Staff	Hanle fees, recorsd, document	Daily	Meduim
IT support	Maintain system , manager accounts and security	As needed	High

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Part 2: Persona Creation

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Create 2–3 personas using the following template:

Persona Name:	Principle	Teacher
Age:	45	32
Education:	Master in business	Bachlero's in math
Role:	Maneger	Teacher
Goals:	To supervise the scholl	Record attendance and grades quickly, share assignments, and spot struggling students early
Motivations:	Can manege and check scoll from one place	Spend less time on paperwork and more time teaching; support student success
Pain Points:	Need to follow every teacher indvisualy	Paper attendance sheets, re-entering grades in multiple places, slow communication with parents
Technical Skill Level:	Meduim - high	Medium: comfortable with email and basic apps, not advanced tools
Usage Context:	Uses a desktop computer in the office all day; needs role-based access and secure data	Uses laptop in class

Persona Name:	Student	Parent
Age:	15	40
Education:	Student	Diploma in Ai
Role:	Student	Parent
Goals:	Solve assainmgnets and quizees, keep attendance tracking	Track child's academic staute
Motivations:	provides all their academic needs in one place	
Pain Points:	Solves every assinnmgint in indvisual platform	No need to goto scholl for following children academic statue
Technical Skill Level:	Meduim	low: comfortable with email and basic apps, not advanced tools
Usage Context:	Uses a desktop computer from home	Uses or phone form any where

Part 3: Persona to Requirements Mapping

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Map each persona to required system features.

Persona	Feature Needed	Priority
Manager	Student enrollment and record management	High
	Timetable builder with conflict detection	
	Role-based access control and security	
	Reports and analytics dashboard	
Teacher	Digital attendance recording	High
	Gradebook with automatic calculations	High
	Assignment posting and submission tracking	High
	Messaging with parents	Medium
Parents	Real-time view of grades and attendance	High
	Bilingual interface	
Student	Solve assinments	high
	Track attendce	

Part 4: Scenarios & User Stories

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(Teacher)

Scenario: At the start of first period, Sara logs into the system on her laptop and marks attendance for her class in under a minute. After the lesson, she enters quiz scores. The system calculates averages and flags two students whose grades are dropping, and she sends a note to their parents.

As a teacher, I want to record attendance digitally so that I save class time and avoid paperwork errors.

As a teacher, I want to see automatic grade calculations and alerts so that I can identify struggling students early.

(Parent)

Scenario: In the evening, Mohammed opens the school app on his phone. He sees a notification that his son was absent today, checks his daughter's latest exam results, and pays the upcoming term fee online. He then messages the teacher about his son's absence.

As a parent, I want to receive instant notifications about my child's attendance so that I can follow up quickly.

As a parent, I want to pay school fees online so that I avoid visiting the school and never miss a deadline.

(Administrator)

Scenario: Before the new semester, Nora enrolls new students and builds the timetable. The system warns her about a room conflict, which she fixes. At the end of the week, she generates an attendance and performance report for the principal.

As an administrator, I want a timetable tool that detects conflicts so that I can schedule classes without clashes.

As an administrator, I want to generate reports automatically so that I can support decisions and reduce manual work.

Submission:

- Personas (2–3)
- Persona-feature mapping
- User stories

Assessment Criteria:

Criterion	Marks
Persona realism & clarity	30%
Alignment with project idea	30%
Requirements mapping	20%
Scenarios & user stories	20%